

Research article

Extreme rainfall events eliminate the response of greenhouse gas fluxes to hydrological alterations and fertilization in a riparian ecosystem

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ABSTRACT

Riparian ecosystems are essential carbon dioxide (CO₂) sources, which considerably promotes climate warming. However, the other greenhouse gas fluxes (GHGs), such as methane (CH₄) and nitrous oxide (N₂O), in the riparian ecosystems have not been well studied, and it remains unclear whether and how these GHG fluxes respond to extreme weather, fertilization and hydrological alterations associated with reservoir management. Here, we assessed the impacts of hydrological alterations (i.e., flooding frequency) and fertilization (nitrogen and/or phosphorus) induced by human activities (hydroengineering construction and agricultural activities) on GHG fluxes, and further investigated the underlying mechanisms in two contrasting years (normal vs. extreme rainfall years) in a reservoir riparian zone dominated by grasses. The significant combined effects of extreme rainfall events and human activities (hydrological alterations and fertilization) on the GHGs were observed. Continuous flooding reduced CO₂ emissions by 24% but increased CH₄ emissions by ~4 times in a normal rainfall year. In addition, nitrogen fertilization promoted CO₂ emissions by 37%. However, these phenomena were not observed in the year with extreme rainfall events, which made the flooding levels homogeneous across the treatments. Furthermore, we found that CO₂ fluxes were driven by the soil moisture, nutrient content, aboveground biomass, and root carbon content, while CH₄ and N₂O fluxes were merely driven by the soil properties (pH, moisture, and nutrient content). This study provides valuable insights into the crucial role of extreme rainfall events, hydrological alteration, and fertilization in regulating GHG fluxes in riparian ecosystems, as well as supports the integration of these changes in GHG emission models.

1. Introduction

Over the past century, reservoirs have expanded globally (Soued et al., 2022), not only providing services such as irrigation, flood control, and hydropower but also creating novel riparian ecosystems where submergence occurs in winter rather than in summer. These riparian

ecosystems are hotspots of carbon dioxide (CO₂) emissions, contributing approximately 26.2 Tg CO₂ year⁻¹ globally (Keller et al., 2021). Besides CO₂, methane (CH₄) and nitrous oxide (N₂O) are other potent greenhouse gases (GHGs), and their fluxes in riparian ecosystems are not well understood (IPCC et al., 2023; Keller et al., 2021). The scarcity of studies can be attributed to the small area of reservoir riparian zone and

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perceived insignificance of GHG emissions. These GHGs are vulnerable to anthropogenic activities and climate change. For instance, hydrological alterations caused by hydroengineering considerably affect the GHG fluxes owing to the changes in soil moisture (Liu and Greaver, 2009; Poblador et al., 2017). Agricultural fertilization also greatly affects the GHG fluxes owing to the changes in nutrient availability (Butterbach-Bahl et al., 2013; Shen and Zhu, 2022). Furthermore, climate change is increasing the incidence of extreme weather events (Slater and Villarini, 2016), which could also impact GHG fluxes by changing soil moisture. However, few studies have focused on this effect in the riparian ecosystems. More importantly, the interactions between extreme weather, hydrological alterations, and fertilization on GHG fluxes are unknown, leading to uncertainty in projecting GHG fluxes in riparian ecosystems under future scenarios and their role in regulating climate change.

In contrast to natural coastal areas and floodplains where flooding is episodic, the reservoir riparian zones are subjected to hydrological alterations (i.e., changes in flooding frequency) that can be regulated by reservoir operation. Flooding generally promotes anaerobic conditions, which reduce carbon mineralization and CO₂ production (Altor and Mitsch, 2008) but benefits methanogenesis and denitrification, two biochemical processes for CH₄ and N₂O production, respectively (Gebremichael et al., 2017; Olefeldt et al., 2017). In addition, the impact of flooding on GHG fluxes depends on flooding frequency. Frequent flooding increases N₂O and CH₄ emissions (Cowan et al., 2021), whereas continuous flooding decreases N₂O emissions in riparian ecosystems owing to the complete denitrification process (Jacinthe et al., 2012; Shi et al., 2021). Considering that extreme weather will frequently occur in the future (IPCC et al., 2023), whether and how extreme rainfall events and hydrological alterations interact to impact GHG fluxes need to be well understood.

Riparian ecosystems are also subjected to nitrogen (N) and phosphorus (P) enrichment as a function of agricultural fertilization (Lyu et al., 2021; Weihrach and Weber, 2020). N and P additions provide nutrients for plant growth and biochemical processes (i.e., denitrification, nitrification, and methanogenesis), thus promoting GHG emissions (Chen et al., 2019, 2020; Xu et al., 2020). However, many studies focus on forests, grasslands, wetlands, and agricultural ecosystems, while limited attention has been paid to riparian ecosystems (He et al., 2021; Shi et al., 2021). In addition, it remains unclear whether the effects of nutrient addition on GHG fluxes are regulated by hydrological alteration and extreme rainfall events.

The objective of this study is to investigate the interactions of extreme rainfall events, hydrological alterations, and fertilization on GHG fluxes in a riparian ecosystem, as well as explore the related mechanisms. We hypothesized that (1) extreme rainfall would promote the flooding effect on the GHG fluxes by further increasing soil moisture, and (2) extreme rainfall would decline the effect of nutrient addition on GHG fluxes because of low soil nutrient content.

2. Materials and methods

2.1. Study site and experimental design

This study was conducted in a riparian zone of the Three Gorges Reservoir located at Zhongxian Revegetation station, Chongqing, China (30°26' N, 108°11' E) (Fig. S1). The region has a subtropical monsoon climate with average annual temperature of 17.2 °C and average annual rainfall of 1087 mm (1981–2010) (Cui et al., 2022). Following the completion of the Three Gorges Reservoir project in 2008, the water level in this riparian zone is regulated by reservoir operation and fluctuates between 145 and 175 m a.s.l., up to 30 m. During the wet season (May to September), the reservoir stores water and increases water table level to 175 m. During the dry season (October to April), the reservoir discharges water and decreases the water table level to 145 m. This pattern is opposite to the natural hydrological regime and has

Table 1

The abbreviations and original words.

Original words	Abbreviations	Original words	Abbreviations
Greenhouse gas	GHG	Simpson's heterogeneity index	D
Carbon dioxide	CO ₂	Pielou evenness index	Jew
Methane	CH ₄	Aboveground biomass	AGB
Nitrous oxide	N ₂ O	Belowground biomass	BGB
No flooding	NF	global warming potential	GWP
Periodic flooding	PF	Shannon-Wiener heterogeneity index	H
Continuous flooding	CF	Total nitrogen	TN
Control without nutrient addition	CK	Total carbon	TC
N addition	N	Total organic carbon	TOC
P addition	P	Soil water content	SWC
combined N and P addition	N + P	Soil temperature	ST
Root carbon content	rootC	Root nitrogen content	rootN

become a worldwide phenomenon in large reservoirs, such as the Grand Coulee (US), Nagarjuna Sagar (India), and Kurobe (Japan) (World Register of Dams, 2020). The soil in the riparian zone is classified as Regosol in the FAO taxonomy system. The vegetation was dominated by grasses (*Cynodon dactylon*, *Physalis minima*, *Polygonum hydropiper*, *Xanthium strumarium*, and *Echinochloa crusgalli*). The abbreviations used in this study were shown in Table 1, and the technical roadmap was shown in Fig. S1.

The manipulated flooding frequency and fertilization were established in May 2019 (Fig. S2). There were three flooding frequencies: (1) control without water addition (no flooding, NF); (2) six days of submergence 5 cm above the soil surface, followed by a six-day well-drained condition (periodic flooding, PF); and (3) continuous submergence 5 cm above the soil surface (continuous flooding, CF). The water table depth was controlled by a pump, and we checked the water level every day to make sure it was under the certain depth. There were four nutrient addition treatments: (1) control without nutrient addition (CK); (2) N addition (urea, 100 kg N ha⁻¹ y⁻¹); (3) P addition (P₄O₁₀, 20 kg P ha⁻¹ y⁻¹); and (4) combined N (urea, 100 kg N ha⁻¹ y⁻¹) and P addition (P₄O₁₀, 20 kg P ha⁻¹ y⁻¹) (N + P). Nutrient addition was divided into three portions and applied to the soil monthly during the growing season from June to August. The amount of nutrient addition was based on the agricultural application rate in this area reported by the government, with the rate of 92 kg N ha⁻¹ y⁻¹ and 7 kg P₂O₅ ha⁻¹ y⁻¹. Three replicate blocks (15 m × 3 m) were selected in the field. Each block was divided into three sub-blocks for the three flooding treatments. Wooden boards were inserted into the soil for more than 50 cm depth for each sub-block, and each sub-block was divided into four plots (1 m × 3 m) for the four nutrient addition treatments.

An extreme rainfall event (377 mm) occurred in July 2020 (Fig. S3), which is defined as the rainfall amount above 95% over ~40 years (Jones et al., 1999). Two extreme rainfall events occurred from 1985 to 2021 (Fig. S1). The 2020 extreme rainfall event provided an opportunity to investigate the impacts of extreme weather on the response of GHGs to different flooding frequencies and nutrient additions in reservoir riparian ecosystems.

2.2. Soil sampling and properties

Soil samples were collected biweekly during the growing season (June–August) from 2019 to 2020. In each collection, five soil cores were collected from the 0–20 cm soil layer in every plot with a 5 cm diameter stainless steel corer after removing the litter. They were divided into two depths, 0–10 cm (topsoil) and 10–20 cm (subsoil), and were thoroughly mixed to form one sample at each depth. Concurrent with soil sampling, the soil temperature (ST) at 5 cm depth was

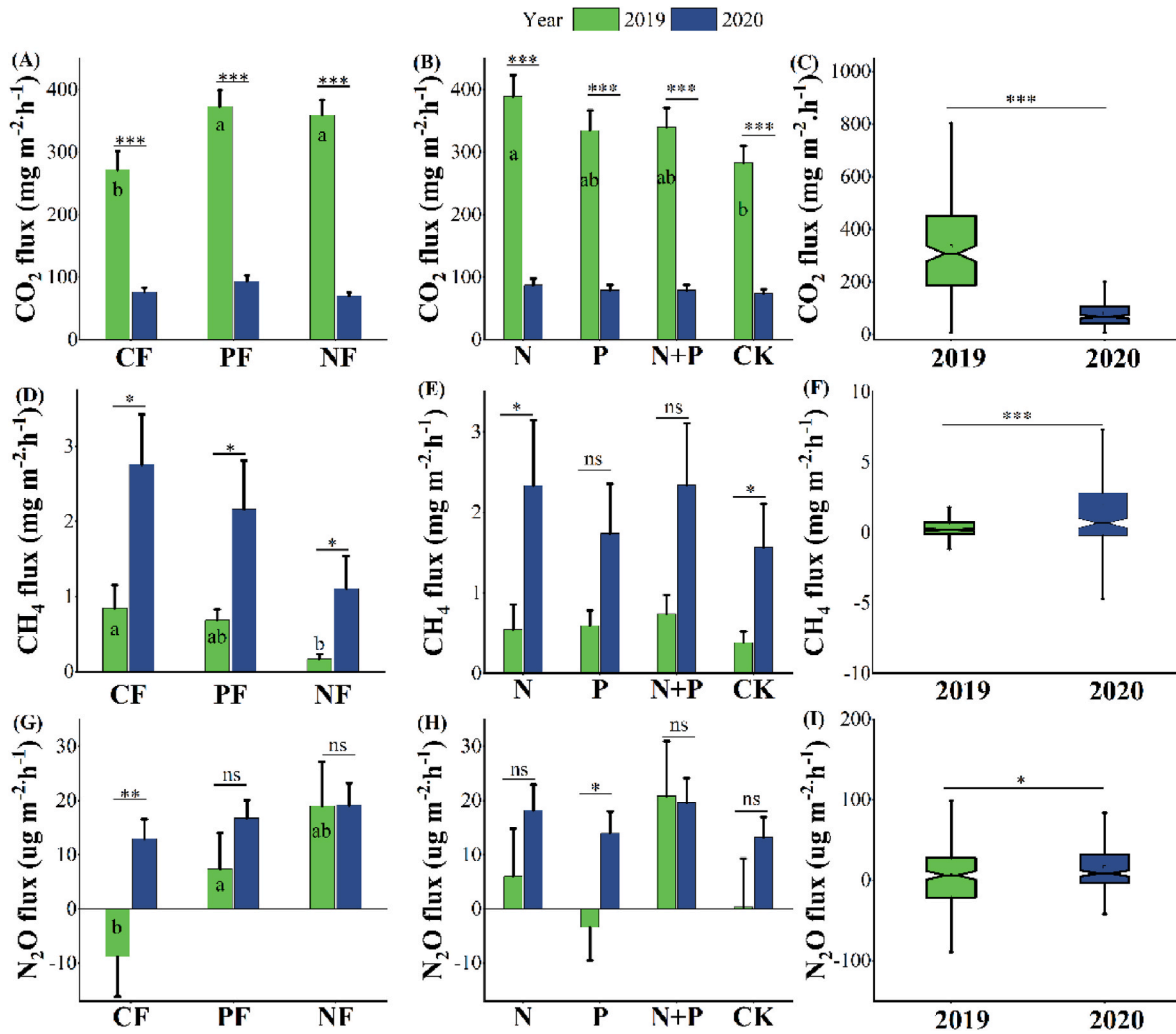


Fig. 1. The GHG fluxes under different flooding frequencies, nutrient addition, and years. “*”, “**”, and “***” represent significance between years at the level of 0.05, 0.01, and 0.001, respectively. Different letters represent the significant difference among treatments in the same year. “ns” represents insignificant difference between years.

measured by a thermometer.

Soil water content (SWC) was measured by drying at 105 °C for 2 days. The soil pH was measured using a pH meter with a soil-to-water ratio of 2.5. The total nitrogen (TN), total carbon (TC), and total organic carbon (TOC) contents were measured by a stable isotope mass spectrometer (Thermo Scientific GmbH, Bremen, Germany). The concentration of dissolved organic carbon (DOC) was determined by extracting 10 g fresh soil with 50 mL of deionized water and filtering (0.45 µm, Waterman, Kent, UK), and analyzed with TOC analyzer (Vario TOC cube, Hanau, Germany). $\text{NH}_4^+\text{-N}$ and $\text{NO}_3^+\text{-N}$ were extracted from the soils using KCl (2M) extraction and determined by a flow auto-analyzer (EasyChem plus, Syntex, Italy).

2.3. Vegetation characteristics

In August 2019–2020, the vegetation diversity in the quadrats (1 m × 1 m) was surveyed in each subplot and assessed using the Shannon-Wiener heterogeneity index (H) (Shannon and Weaver, 1949), Simpson's heterogeneity index (D) (Simpson, 1949), and Pielou evenness index (Jew) (Pielou, 1966). Aboveground biomass (AGB) was weighed after clipping and drying at 70 °C for 48 h. Plant roots were sampled using a root corer (16 cm in diameter and 20 cm in length). Belowground biomass (BGB) was determined gravimetrically after drying the roots at 70 °C for 48 h (Hefting et al., 2005). The carbon and nitrogen content of

the roots was measured by a stable isotope mass spectrometer (Thermo Scientific GmbH, Bremen, Germany).

2.4. Greenhouse gas measurement and global warming potential

The static chamber method was used to sample air for greenhouse gas (CO_2 , CH_4 , and N_2O) measurements (Crill et al., 1988; Shi et al., 2021). Gas samples were taken from opaque chambers (50 cm in height and 34.8 cm in diameter) fitted to the groove of polyvinyl chloride biweekly during the growing seasons (June–August) of 2019–2020. There were two times each month for the measurement. Vegetation was not removed in the chamber when measuring GHGs; therefore, the CO_2 fluxes measured in this study represented ecosystem respiration. A fan was installed in each chamber to mix the air during sampling, and a thermometer was placed in the chamber to measure the atmospheric temperature. Gas samples were collected from the chamber at 10 min-intervals from 0 to 30 min using plastic syringes. Gas samples were stored in gas sampling bags (Hede Tech Inc., Dalian, Shandong, China) and subsequently analyzed by gas chromatography (Penkin Elmer Inc., MA, USA) within 24 h. GHG fluxes were calculated as follows:

$$F = \frac{dc}{dt} \cdot \rho \cdot \frac{P}{P_0} \cdot \frac{T_0}{T} \cdot \frac{V}{A} \quad (1)$$

Table 2

Extreme rainfall event impacted the response of the soil and vegetation characteristics to flooding frequency.

		2019 (normal rainfall)			2020 (Extreme rainfall)		
		CF	PF	NF	CF	PF	NF
Soil (0–10 cm)	ST (°C)	26.12 ± 0.19Ab	26.89 ± 0.22Aa	27.01 ± 0.24Aa	24.95 ± 0.16Bb	25.49 ± 0.21Ba	25.75 ± 0.26Ba
	pH	6.66 ± 0.05Ba	6.65 ± 0.05Ba	6.21 ± 0.05Bb	7.27 ± 0.05Aa	7.19 ± 0.05Aa	6.50 ± 0.08Ab
	SWC (%)	34.98 ± 1.13Ba	28.36 ± 1.16Bb	26.69 ± 1.66Bb	38.59 ± 1.03Aa	35.23 ± 1.70Aa	35.27 ± 1.37Aa
	NH ₄ ⁺ (mg/kg)	15.93 ± 1.16Aa	15.53 ± 1.25Aa	13.81 ± 1.04Aa	2.67 ± 0.17Ba	2.58 ± 0.15Ba	2.62 ± 0.17Ba
	NO ₃ ⁻ (mg/kg)	1.87 ± 0.41Ba	1.95 ± 0.31Ba	1.51 ± 0.28Ba	3.46 ± 0.57Aa	3.79 ± 0.53Aa	3.38 ± 0.60Aa
	DOC (mg/kg)	236.51 ± 27.05Aa	270.80 ± 25.94Aa	260.72 ± 25.61Aa	66.21 ± 7.54Ba	70.17 ± 7.61Ba	87.27 ± 9.30Ba
	TN (%)	0.13 ± 0.01Aa	0.13 ± 0.01Aa	0.15 ± 0.01Aa	0.12 ± 0.01Aa	0.12 ± 0.004Aa	0.119 ± 0.004Ba
	TOC (%)	1.10 ± 0.05Ab	1.15 ± 0.05Aab	1.32 ± 0.07Aa	1.20 ± 0.05Aa	1.22 ± 0.05Aa	1.24 ± 0.04Aa
	TC (%)	1.17 ± 0.04Bb	1.21 ± 0.05Aab	1.37 ± 0.06Aa	1.31 ± 0.05Aa	1.30 ± 0.04Aa	1.29 ± 0.04Aa
	C:N	8.62 ± 0.14Bb	8.78 ± 0.13Bb	9.15 ± 0.11Ba	10.35 ± 0.13Aa	10.32 ± 0.15Aa	10.48 ± 0.18Aa
Soil (10–20 cm)	pH	6.63 ± 0.06Ba	6.28 ± 0.07Bb	6.36 ± 0.06Bb	7.15 ± 0.06Aa	6.96 ± 0.06Aa	6.56 ± 0.07Ab
	SWC (%)	28.64 ± 0.96Aa	26.49 ± 1.01Aa	22.74 ± 1.20Bb	28.84 ± 0.94Aa	27.10 ± 0.71Aa	28.09 ± 0.91Aa
	NH ₄ ⁺ (mg/kg)	7.75 ± 0.84Aa	6.51 ± 0.67Aa	7.27 ± 0.84Aa	3.00 ± 0.22Ba	2.79 ± 0.19Ba	2.86 ± 0.19Ba
	NO ₃ ⁻ (mg/kg)	3.08 ± 0.59Ab	3.60 ± 0.60Aa	3.29 ± 0.57Ab	1.59 ± 0.21Bab	1.65 ± 0.21Ba	1.33 ± 0.19Bb
	DOC (mg/kg)	253.34 ± 29.32Aa	254.23 ± 31.74Aa	253.21 ± 31.58Aa	83.70 ± 8.90Bb	104.25 ± 8.40Ba	109.59 ± 9.46Ba
	TN (%)	0.09 ± 0.003Ab	0.10 ± 0.004Aa	0.10 ± 0.003Aa	0.08 ± 0.003Ba	0.08 ± 0.003Ba	0.08 ± 0.003Ba
	TOC (%)	0.65 ± 0.03Bb	0.81 ± 0.04Aa	0.75 ± 0.03Aab	0.75 ± 0.03Aa	0.77 ± 0.02Aa	0.78 ± 0.03Aa
	TC (%)	0.71 ± 0.03Bb	0.88 ± 0.05Aa	0.84 ± 0.03Aa	0.85 ± 0.05Aa	0.81 ± 0.02Aa	0.84 ± 0.03Aa
	C:N	7.49 ± 0.20Ba	8.06 ± 0.15Ba	7.80 ± 0.19Ba	9.88 ± 0.25Aa	9.64 ± 0.19Aa	9.61 ± 0.19Aa
	AGB (g/m ²)	213.56 ± 56.31Ba	232.72 ± 73.69Ba	218.26 ± 18.69Ba	482.43 ± 51.53Aa	639.83 ± 64.41Aa	483.71 ± 40.77Aa
vegetation	BGB (g/m ²)	496.78 ± 83.05Aa	354.39 ± 53.81Aa	397.68 ± 70.71Aa	677.79 ± 108.40Aa	641.29 ± 130.42Aa	446.48 ± 73.68Aa
	RootC (%)	47.61 ± 1.14Ab	51.86 ± 1.22Aa	48.33 ± 1.75Aab	36.86 ± 0.75Bb	39.25 ± 0.36Ba	37.44 ± 0.36Bb
	RootN (%)	1.71 ± 0.05Aa	1.26 ± 0.05Ab	1.05 ± 0.06Ac	1.00 ± 0.06Bab	1.00 ± 0.05Ba	0.81 ± 0.07Bb
	Root C:N	28.29 ± 1.46Bb	41.93 ± 1.93Aa	48.92 ± 4.25Aa	38.51 ± 2.37Ab	40.43 ± 1.67Aab	51.01 ± 4.67Aa
	H	1.41 ± 0.18Ab	1.72 ± 0.06Bb	2.04 ± 0.05Aa	1.96 ± 0.13Aa	2.03 ± 0.07Aa	1.95 ± 0.02Aa
	D	0.55 ± 0.03Aa	0.49 ± 0.01Aa	0.47 ± 0.003Aa	0.49 ± 0.01Aa	0.48 ± 0.004Aa	0.47 ± 0.003Aa
	Jsw	0.98 ± 0.01Aa	0.97 ± 0.003Aa	0.98 ± 0.004Aa	0.95 ± 0.01Aa	0.96 ± 0.01Aa	0.97 ± 0.01Aa

Note: “PF” represents periodic flooding, “CF” represents continuous flooding, “NF” represents not flooding. The different lowercase letters represent the significant difference between treatments in the same year, while the caption letters represent the significant difference between two years.

where F is the GHG fluxes ($\text{mg m}^{-2} \text{h}^{-1}$), $\frac{dc}{dt}$ is the change in concentration over time during 30 min of sampling (ppm s^{-1}); ρ represents the gas density (mg cm^{-3}); P and T are the atmospheric pressure (Pa) and air temperature (K) in the chamber, respectively; P_0 and T_0 are the atmospheric pressure (Pa) and air temperature (K) under standard conditions, respectively; V is the volume of the chamber (m^3); and A is the area covered by the chamber (m^2). Fluxes were accepted when $R^2 \geq 0.8$ (evaluating the regression between gas concentration and interval time). Therefore, approximately 7% of the data were excluded. The global warming potential (GWP) over a 100-year time horizon was calculated by summing the cumulative CO_2 multiplied by 1, cumulative CH_4 multiplied by 27, and cumulative N_2O multiplied by 273 (IPCC et al., 2023). We calculated GWP from June to August in 2019 and 2020, instead of annually because GHG fluxes were measured during these months with water table drawdown and exposure of riparian zone. The mean value of GHG fluxes under each treatment and linear interpolation were applied to extrapolate these samples to represent the entire month. There were about 72 samples each month, and the margin of error was the standard error calculated from those samples.

2.5. Statistical analysis

A linear mixed-effects model was used to investigate the impacts of extreme rainfall events, flooding frequency, and nutrient addition on three GHG fluxes, soil (ST, SWC, pH, TOC, TC, TN, DOC, NH_4^+ , and NO_3^-), and vegetation properties (H, D, Jsw, AGB, BGB, and root carbon and nitrogen content). The block was considered as a random factor. Flooding frequency, nutrient addition, and year were considered as fixed factors. Tukey's post-hoc test was used to determine the differences among treatments. The impacts of extreme rainfall on GHG fluxes under different treatments were analyzed using a paired t -test. A quantile-quantile plot (QQ plot) was used for the normality test (Fig. S4), and data were log-transformed when necessary before analysis. Pearson's correlation was used to analyze the relationships among GHG fluxes, soil, and vegetation properties. A stepwise regression analysis was

applied to identify the important drivers of GHG fluxes, and their relative importance was analyzed by ‘relaimpo’ package. These drivers were used in structural equation modeling (SEM) to investigate their direct and indirect effects on GHG fluxes. Fisher's C-test was used to confirm the accuracy of the modeling results. However, there was only one driver for the CH_4 and N_2O fluxes based on stepwise regression analysis. Consequently, we selected other drivers for the SEM according to the Pearson correlation, and the variance inflation factor ($\text{VIF} < 5$) was used to avoid multiple collinearities among the drivers. These analyses were performed in R version 4.0.3 (R Core Team, 2018) using ‘piecewiseSEM’, ‘lmerTest’, and ‘lsmeans’ packages.

3. Results

3.1. GHG fluxes

There were significantly combined effects of extreme rainfall events and treatments on the GHG fluxes. During the normal rainfall year (2019), flooding frequency and nutrient addition significantly affected GHG fluxes (Fig. 1 and S5). The effects of flooding frequency on the GHG fluxes were also observed in 2021, the other normal rainfall year (Fig. S6). Continuous flooding (CF) led to a substantial reduction in CO_2 emissions by 24% ($p < 0.001$) compared with the no flooding treatment (NF), while simultaneously increasing CH_4 emissions by ~ 4 times ($p < 0.05$). Periodic flooding (PF) significantly increased N_2O emissions by ~ 3 times compared to CF ($p < 0.05$). Furthermore, N addition resulted in a notable increase in CO_2 emissions by 37% ($p < 0.05$). However, there were no significant effects of flooding frequency or nutrient addition on GHG fluxes in extreme rainfall year.

Compared with 2019, the extreme rainfall year (2020) significantly reduced CO_2 emissions by $\sim 75\%$ ($p < 0.001$) and increased CH_4 by ~ 3 times under all flooding treatments ($p < 0.05$) (Table S1). Extreme rainfall event also increased N_2O fluxes by 2.5 times under the CF treatment ($p < 0.01$) (Fig. 1). In addition, the extreme rainfall event significantly reduced CO_2 emission by $\sim 76\%$ under all nutrient addition

Table 3

Extreme rainfall event impacted the response of the soil and vegetation characteristics to nutrient enrichment.

		2019 (normal rainfall)				2020 (Extreme rainfall event)			
		N	P	N + P	CK	N	P	N + P	CK
Soil (0–10 cm)	ST (°C)	26.53 ± 0.26Aa	26.70 ± 0.26Aa	26.73 ± 0.250Aa	26.74 ± 0.26Aa	25.30 ± 0.24Ba	25.32 ± 0.22Ba	25.49 ± 0.25Ba	25.49 ± 0.28Ba
	pH	6.49 ± 0.07Ba	6.45 ± 0.08Ba	6.53 ± 0.07Ba	6.56 ± 0.07Ba	6.97 ± 0.10Aa	7.00 ± 0.08Aa	6.96 ± 0.09Aa	7.01 ± 0.10Aa
	SWC (%)	29.29 ± 1.63Ba	29.39 ± 1.56Ba	29.88 ± 1.47Ba	31.48 ± 2.03Ba	37.61 ± 1.76Aa	34.24 ± 1.44Aa	36.54 ± 1.63Aa	37.04 ± 1.62Aa
	NH ₄ ⁺ (mg/kg)	16.18 ± 1.47Aa	12.87 ± 1.14Aa	16.46 ± 1.41Aa	14.82 ± 2.21Aa	2.73 ± 0.23Ba	2.67 ± 0.17Ba	2.56 ± 0.10Ba	2.54 ± 0.21Ba
	NO ₃ ⁻ (mg/kg)	1.59 ± 0.21Ba	1.43 ± 0.28Ba	1.96 ± 0.38Ba	2.11 ± 0.57Aa	3.17 ± 0.61Aa	3.59 ± 0.66Aa	3.99 ± 0.65Aa	3.43 ± 0.70Aa
	DOC (mg/kg)	244.43 ± 30.25Aa	249.57 ± 29.54Aa	265.92 ± 31.24Aa	265.39 ± 30.31Aa	81.59 ± 10.36Ba	79.67 ± 9.67Ba	66.35 ± 9.11Ba	71.28 ± 9.02Ba
	TN (%)	0.13 ± 0.01Aa	0.14 ± 0.010Aa	0.13 ± 0.01Aa	0.14 ± 0.01Aa	0.12 ± 0.006Aa	0.11 ± 0.004Ba	0.12 ± 0.004Aa	0.12 ± 0.005Ba
	TOC (%)	1.13 ± 0.07Aa	1.20 ± 0.08Aa	1.19 ± 0.05Aa	1.24 ± 0.07Aa	1.25 ± 0.07Aa	1.17 ± 0.04Aa	1.22 ± 0.05Aa	1.24 ± 0.05Aa
	TC (%)	1.26 ± 0.07Aa	1.22 ± 0.04Aa	1.24 ± 0.06Aa	1.30 ± 0.07Aa	1.33 ± 0.07Aa	1.26 ± 0.05Aa	1.32 ± 0.05Aa	1.30 ± 0.05Aa
	C:N	8.69 ± 0.16Ba	8.71 ± 0.14Ba	8.95 ± 0.14Ba	9.06 ± 0.16Ba	10.44 ± 0.17Aa	10.34 ± 0.19Aa	10.21 ± 0.16Aa	10.55 ± 0.18Aa
Soil (10–20 cm)	pH	6.43 ± 0.08Ba	6.37 ± 0.08Ba	6.41 ± 0.07Ba	6.47 ± 0.08Ba	6.90 ± 0.09Aa	6.91 ± 0.08Aa	6.87 ± 0.09Aa	6.89 ± 0.08Aa
	SWC (%)	24.52 ± 1.00Ba	26.55 ± 1.43Aa	25.63 ± 1.39Aa	27.13 ± 1.34Aa	28.21 ± 1.05Aa	28.62 ± 0.97Aa	26.45 ± 0.91Aa	28.76 ± 1.01Aa
	NH ₄ ⁺ (mg/kg)	7.60 ± 0.71Aa	7.03 ± 0.96Aa	8.15 ± 1.24Aa	5.92 ± 0.55Aa	2.78 ± 0.21Ba	2.96 ± 0.22Ba	2.82 ± 0.23Ba	2.97 ± 0.27Ba
	NO ₃ ⁻ (mg/kg)	3.17 ± 0.69Aa	3.60 ± 0.74Aa	3.41 ± 0.68Aa	3.12 ± 0.60Aa	1.37 ± 0.26Ba	1.55 ± 0.21Ba	1.55 ± 0.23Ba	1.62 ± 0.23Ba
	DOC (mg/kg)	240.20 ± 34.53Aa	260.92 ± 35.63Aa	277.54 ± 38.38Aa	235.72 ± 33.40Aa	92.55 ± 9.89Ba	100.30 ± 11.26Ba	96.50 ± 9.75Ba	107.37 ± 10.76Ba
	TN (%)	0.09 ± 0.002Aa	0.09 ± 0.004Aa	0.10 ± 0.004Aa	0.10 ± 0.01Aa	0.08 ± 0.003Ba	0.08 ± 0.003Ba	0.08 ± 0.003Ba	0.08 ± 0.004Ba
	TOC (%)	0.68 ± 0.02Aa	0.72 ± 0.03Ba	0.77 ± 0.05Aa	0.79 ± 0.05Aa	0.73 ± 0.03Aa	0.82 ± 0.03Aa	0.74 ± 0.02Aa	0.78 ± 0.04Aa
	TC (%)	0.75 ± 0.03Aa	0.80 ± 0.03Aa	0.83 ± 0.05Aa	0.86 ± 0.05Aa	0.83 ± 0.07Aa	0.88 ± 0.03Aa	0.79 ± 0.02Aa	0.87 ± 0.04Aa
	C:N	7.69 ± 0.15Ba	7.62 ± 0.27Ba	7.97 ± 0.17Ba	7.85 ± 0.23Ba	9.72 ± 0.29Aa	9.72 ± 0.20Aa	9.83 ± 0.23Aa	9.56 ± 0.25Aa
vegetation	AGB (g/m ²)	nd	nd	nd	nd	542.98 ± 52.49a	582.86 ± 90.03a	557.88 ± 54.69Aa	457.56 ± 53.59Aa
	BGB (g/m ²)	434.32 ± 90.78Aa	403.19 ± 58.36Aa	449.60 ± 85.18Aa	378.01 ± 92.91Aa	494.02 ± 72.46Aa	547.77 ± 141.93Aa	707.64 ± 80.55Aa	604.65 ± 175.21Aa
	RootC (%)	48.79 ± 1.31Aa	50.79 ± 1.58Aa	49.83 ± 1.98Aa	47.66 ± 1.79Aa	38.52 ± 0.55Ba	37.88 ± 0.64Ba	37.49 ± 0.21Ba	37.51 ± 1.05Ba
	RootN (%)	1.38 ± 0.10Aa	1.33 ± 0.13Aa	1.29 ± 0.11Aa	1.35 ± 0.10Aa	1.06 ± 0.07Ba	0.94 ± 0.09Bab	0.90 ± 0.06Bab	0.83 ± 0.06Bb
	Root C:N	37.56 ± 3.52Aa	42.53 ± 5.35Aa	42.10 ± 4.88Aa	36.66 ± 2.68Aa	38.24 ± 2.95Aa	43.55 ± 4.09Aa	43.75 ± 3.47Aa	47.72 ± 4.97Aa
	H	1.89 ± 0.01Aa	1.62 ± 0.23Aa	1.76 ± 0.17Aa	1.62 ± 0.24Aa	1.86 ± 0.04Aa	2.19 ± 0.11Aa	1.96 ± 0.02Aa	1.91 ± 0.07Aa
	D	0.48 ± 0.004Aa	0.52 ± 0.04Aa	0.50 ± 0.02Aa	0.52 ± 0.04Aa	0.49 ± 0.01Aa	0.48 ± 0.01Aa	0.47 ± 0.001Aa	0.48 ± 0.004Aa
	Jew	0.97 ± 0.007Aa	0.98 ± 0.01Aa	0.97 ± 0.01Aa	0.98 ± 0.002Aa	0.94 ± 0.02Aa	0.98 ± 0.01Aa	0.96 ± 0.01Aa	0.96 ± 0.004Aa

Note: “N” represents nitrogen addition, “P” represents phosphorus addition, “N + P” represents nitrogen and phosphorus addition, “CK” represents control without nutrient addition. “nd” represents not measured for the treatments. The different lowercase letters represent the significant difference between treatments in the same year, while the caption letters represent the significant difference between two years.

treatments ($p < 0.001$), increased CH₄ emissions by ~3 times under the N addition and CK treatments ($p < 0.05$), and increased N₂O emissions by 5 times under the P addition treatment ($p < 0.05$) (Fig. 1).

3.2. Soil properties

Flooding frequency significantly affected the soil properties during normal and extreme rainfall years, whereas nutrient addition did not impact soil properties during both years (Tables 2 and 3). In normal rainfall year, compared to NF, both CF and PF increased the pH by ~7% ($p < 0.001$) and SWC by 31% and 6% ($p < 0.001$), respectively. Moreover, CF decreased ST by 3% ($p < 0.001$), TOC by 17% ($p < 0.05$), TN by 12% ($p < 0.05$), and TC content by 15% ($p < 0.05$), while PF increased NO₃⁻ content by 29% ($p < 0.001$). In extreme rainfall year, compared to NF, both CF and PF increased soil pH by ~11% ($p < 0.001$). Moreover, CF decreased the DOC concentration by 24% ($p < 0.01$), whereas PF increased the NO₃⁻ content by 24% ($p < 0.01$).

Compared to the normal rainfall year, the extreme rainfall year significantly increased soil pH by ~7% ($p < 0.001$), decreased NH₄⁺ by

~83% ($p < 0.001$), and DOC concentration by ~70% under different flooding frequencies ($p < 0.001$) (Table 2). Similarly, extreme rainfall year significantly increased soil pH by 7% ($p < 0.001$) and decreased NH₄⁺ by ~82% ($p < 0.001$), DOC by ~70% ($p < 0.001$), and TN content by ~14% ($p < 0.01$) under nutrient addition (Table 3).

3.3. Vegetation characteristics

Flooding frequency significantly affected vegetation characteristics during normal and extreme rainfall years, whereas nutrient addition had a limited effect (Tables 2 and 3). Compared with NF, CF and PF increased the root N content in normal rainfall year by 64% ($p < 0.001$) and 21% ($p < 0.05$), respectively, but decreased vegetation diversity (H) by 31% ($p < 0.05$) and 15% ($p < 0.05$), respectively. In the extreme rainfall year, CF decreased the root C:N ratio by 24% ($p < 0.05$), whereas PF increased the root C and N contents by 5% ($p < 0.01$) and 23% ($p < 0.05$), respectively. In addition, only N addition increased root N content by 26% during extreme rainfall year ($p < 0.05$).

Compared with normal rainfall year, extreme rainfall year also

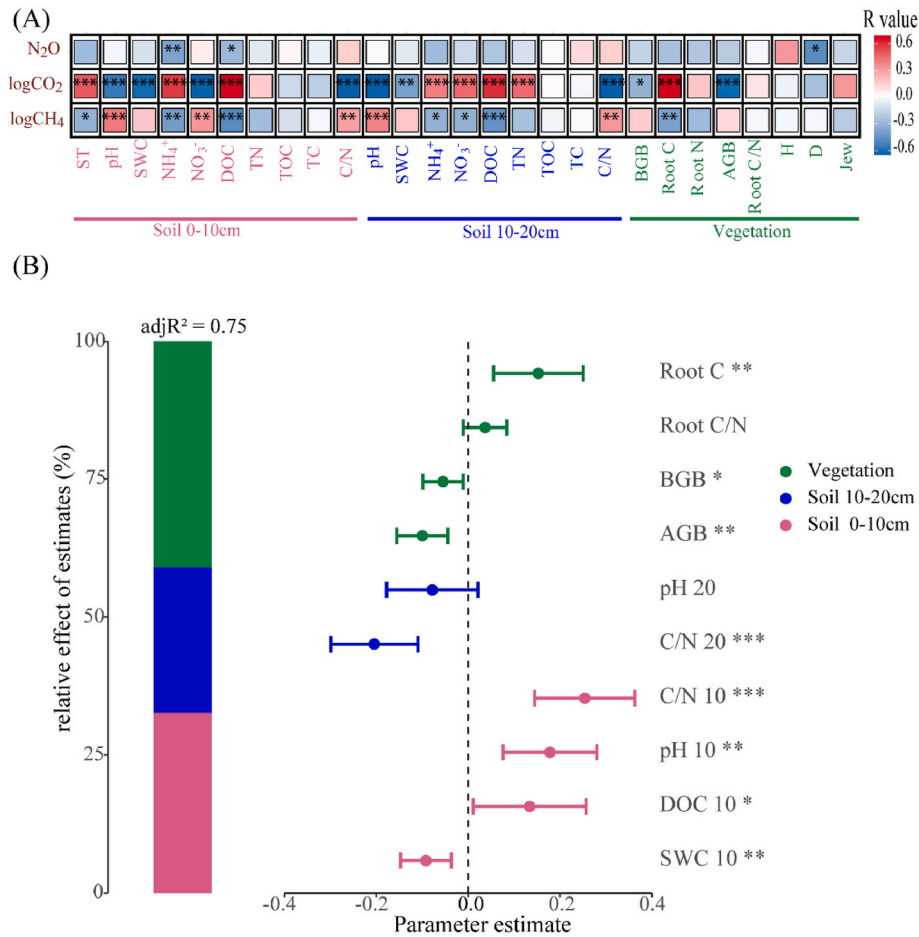


Fig. 2. (A) The relationships between GHG fluxes and characteristics of soil and vegetation. (B) The controls for the CO₂ fluxes according to the stepwise regression analysis. “*”, “**”, and “***” represent significance among treatments and years at the level of 0.05, 0.01, and 0.001, respectively. The number after the soil properties represents the soil depth.

significantly increased AGB by ~1.5 times ($p < 0.01$) and decreased root C ($p < 0.001$) and N ($p < 0.01$) by ~22% under different flooding frequencies (Table 2). In addition, extreme rainfall year reduced the root C ($p < 0.001$) and N ($p < 0.01$) content by 21%–39% under nutrient addition (Table 3).

3.4. Main controls of GHG fluxes

Significant relationships existed between the GHG fluxes and the characteristics of soil and vegetation (Fig. 2). According to stepwise regression analysis, soil properties (pH, SWC, DOC, and C:N ratio) and vegetation characteristics (AGB, BGB, root C, and N content) explained 75% of the variation in CO₂ fluxes (Table S2). Soil properties accounted for ~60% of the explanation. Furthermore, pH and NH₄⁺ were essential drivers of CH₄ and N₂O fluxes, respectively (Table S2). pH explained 19% of the variation in CH₄ fluxes, while NH₄⁺ content explained 10% of the variation in N₂O fluxes (Table S2).

Based on SEM analysis, extreme rainfall and flooding frequency affected CO₂ fluxes by altering soil properties (SWC, pH, DOC, and C:N ratio) and vegetation characteristics (AGB, BGB, root C content, and root C:N ratio) (Fig. 3). In terms of CH₄, extreme rainfall affected CH₄ fluxes by changing soil properties (SWC, pH, and DOC) and root C content, whereas flooding frequency affected CH₄ fluxes only by changing soil properties. In terms of N₂O, extreme rainfall and flooding frequency affected N₂O fluxes by altering soil properties (SWC, DOC, TC, TOC, and NH₄⁺ content). Compared to the flooding frequency, the total effects of extreme rainfall on CO₂, CH₄, and N₂O fluxes were ~3, ~1, and ~5

times higher, respectively.

3.5. Global warming potential

The extreme rainfall year modulated the response of the GWP to different flooding frequencies and nutrient addition (Fig. 4). In the normal rainfall year, CF significantly decreased the GWP by 31% ($P < 0.01$), whereas N and N + P significantly increased the GWP by 39% and 32% ($P < 0.05$), respectively. These effects were not observed during extreme rainfall year. In addition, compared with normal rainfall year, extreme rainfall year significantly decreased GWP by 40–72% ($P < 0.001$) under different flooding frequencies and nutrient additions, and potentially shifted the dominant GHGs from CO₂ to CH₄ (Fig. 4). The CO₂ accounted for ~90% of GWP in normal rainfall year, while it accounted for ~50% of GWP in extreme rainfall year. The CH₄ accounted for ~5% of GWP in normal rainfall year, while it accounted for ~40% of GWP in extreme rainfall year.

4. Discussion

Reservoir riparian zones are considered as important GHG sources (Keller et al., 2021). However, with the increase of extreme weather events and anthropogenic activities, such as hydroengineering construction and fertilization, there is a large uncertainty in predicting GHG fluxes in riparian ecosystems. Moreover, the unique geomorphological features of riparian ecosystems and their high sensitivity to rainfall events result in an urgent need to have a deep understanding in this area

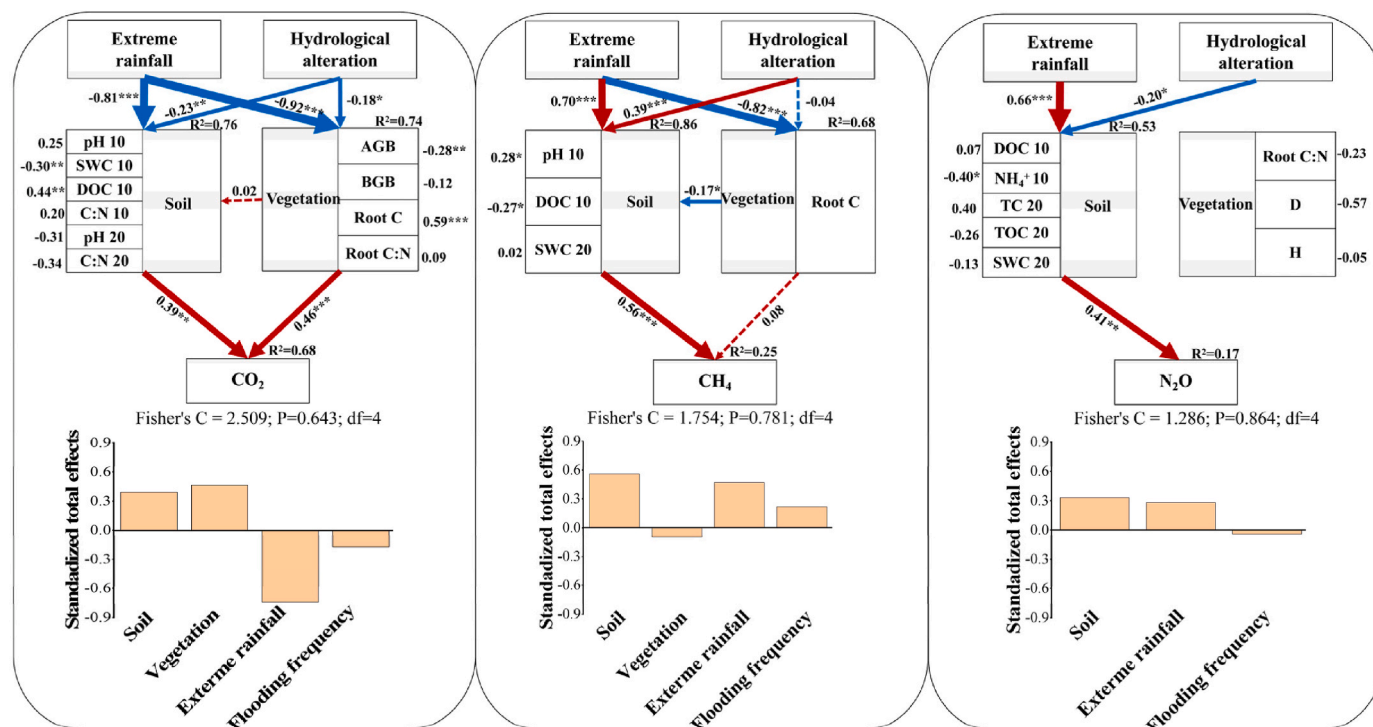


Fig. 3. The direct and indirect relationships among extreme rainfall, flooding frequency, soil properties, vegetation characteristics, and GHG fluxes.

(Pinto et al., 2023). This study aimed to fill the knowledge gap and revealed that the impacts of these changes on GHG fluxes are not merely additive. The extreme rainfall weather can override the responses of GHGs to different flooding frequencies and fertilization. Given a growing number of hydroengineering projects and expanding riparian zones globally, this study is crucial for accurately predicting GHG fluxes in riparian ecosystems under future scenarios.

4.1. Extreme rainfall eliminated the effects of flooding frequency on CO₂ and CH₄ fluxes

The first year of our study was characterized by a normal rainfall regime. In this year, compared with NF, CF decreased CO₂ emissions in the riparian ecosystem owing to its significant effects on soil properties and vegetation characteristics (Fig. 3). On the one hand, high soil moisture and low nutrient content (C:N, TOC, and TN content) caused by CF reduce microbial activity and respiration (Xu et al., 2021) (Table 2). On the other hand, CF significantly reduced root carbon content, which is closely related to metabolic processes and root respiration (Table 2) (Hasibeder et al., 2015). This result aligns with previous studies showing that flooding reduces CO₂ emissions in wetlands and agricultural ecosystems owing to low respiration and diffusion under high soil moisture conditions (Franco-Luesma et al., 2020; Olefeldt et al., 2017; Petrakis et al., 2017). However, it remains uncertain whether flooding reduces CO₂ emissions mainly by decreasing microbial respiration or plant respiration. Plant respiration is related to vegetation characteristics (AGB and root carbon content), while microbial respiration and gas diffusion are related to the soil properties (DOC and SWC). Our SEM analysis revealed that the total effect of vegetation characteristics on CO₂ variation (0.47) was slightly larger than that of soil properties (0.39) under different flooding frequencies (Fig. 3). Therefore, we can assume that flooding decreases plant respiration to a greater extent than microbial respiration in riparian ecosystems. Interestingly, the negative effect of CF on CO₂ emissions disappeared in the extreme rainfall year, which is contrary to our first hypothesis and can be owing to the overshadowing impact of extreme rainfall on soil properties and vegetation characteristics (Fig. 3, Table 2). Specifically, extreme rainfall decreased

soil DOC by ~66%, decreased root carbon content by 23%, and increased AGB by 1.5 times, while CF had a limited effect on them (Table 2). Lower soil DOC can reduce the carbon available for microbial respiration (Dalmagro et al., 2018). Lower root carbon content can reduce metabolic processes and root respiration (Hasibeder et al., 2015). A higher AGB indicates an alteration in carbon allocation under extreme rainfall events, with more carbon for plant growth and less carbon for plant respiration (Caudle and Maricle, 2012).

While CF reduced CO₂ emissions compared to NF, it promoted other potent GHG-CH₄ emissions in normal rainfall year (Fig. 1). The result is in accordance with previous studies conducted in wetlands, forests, and agricultural ecosystems, where anoxic conditions favor methanogenesis and CH₄ production (Sapkota et al., 2020; Schindler et al., 2020; Xu et al., 2021). In addition to the high soil moisture, CF also affected CH₄ emissions by increasing soil pH (Table 2; Fig. 3), which stimulates microbial activity and promotes CH₄ production (Takakai et al., 2021). However, the positive effect of CF on CH₄ fluxes disappeared in the extreme rainfall year (Fig. 1), which can be attributed to the reduction in available carbon for CH₄ production during the extreme rainfall year. In addition, different from the previous study demonstrated the essential role of soil temperature played in CH₄ fluxes (Gomez-Casanovas et al., 2020), we found that pH was the dominant driver. This result can be attributed to the narrow range of temperature in our study site, which merely ranged from 22.3 to 29.9 °C.

CF did not significantly impact N₂O fluxes compared to NF during the normal rainfall year. A possible reason is that high soil moisture not only promotes N₂O production but also promotes N₂O consumption via the complete denitrification process. However, this result is incompatible with previous studies conducted in other ecosystems. Flooding has been reported to decrease N₂O emissions in agricultural ecosystems owing to the stimulation of the complete denitrification process (Khalid et al., 2019) but promotes N₂O emissions in wetland and forest ecosystems due to the increase in ammonia oxidation and nitrifier denitrification activity (Bonetti et al., 2022; Poblador et al., 2017). Consequently, it would take riparian ecosystems into account separately in global N₂O models. Furthermore, compared with PF, CF significantly decreased N₂O emissions, which can be attributed to the lower soil

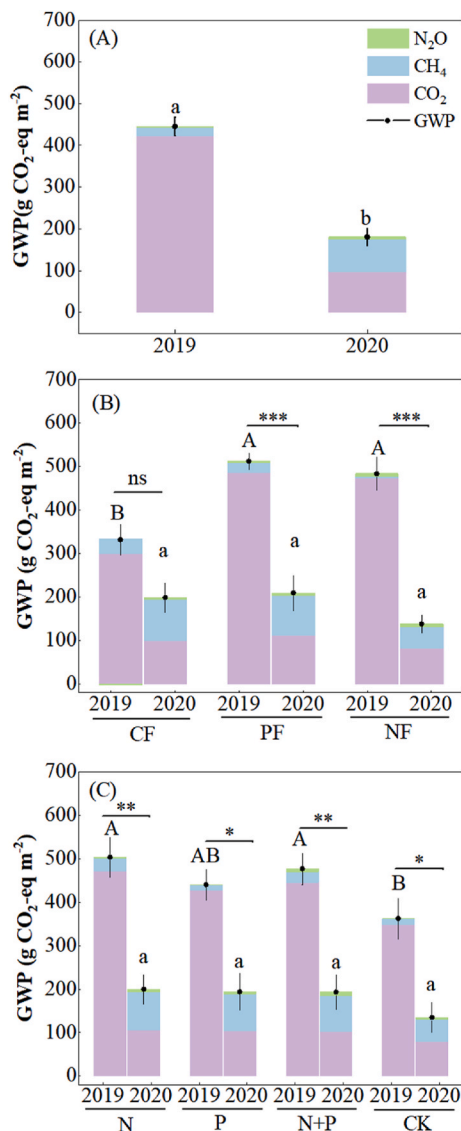


Fig. 4. Global warming potential under different flooding frequencies and nutrient addition in 2019 and 2020. Different capital letters represent the significant difference among treatments in 2019, while different lowercase letters represent the significant difference among treatments in 2020. “***” represents the significant difference between two years.

nutrient content (NO_3^- , TC, and TOC content) for the denitrification process and N_2O production (Table 2) (Ma et al., 2022). However, this CF effect was not observed in the extreme rainfall year, which can be ascribed to the overriding effect of extreme rainfall on soil properties (SWC, TOC, and TC), which are the major drivers of N_2O fluxes (Fig. 3; Table 2). Different from the previous study demonstrated the essential role of soil temperature played in N_2O fluxes (Zhao et al., 2024), we found that available nitrogen was the dominant driver. The different results can be attributed to the two reasons. On the one hand, the narrow range of soil temperature in this study could weaken its effect. On the other hand, the N_2O production processes are driven by nitrogen availability (Zhao et al., 2021); thus, the impact of soil temperature on the N_2O fluxes might be concealed by the available nitrogen.

4.2. Extreme rainfall alleviates the positive effect of nitrogen addition on CO_2 fluxes

Nitrogen addition was found to increase CO_2 fluxes during the normal rainfall year, possibly owing to the increase of belowground

biomass (BGB) that is positively linked to respiration (Table 2) (Wang et al., 2008). This finding is in line with studies conducted in grasslands, wetlands, and agricultural ecosystems, where N addition has been shown to enhance plant growth and respiration (Chen et al., 2020; Deng et al., 2020). However, the positive effect of N addition on CO_2 fluxes was not observed during extreme rainfall year, which supports the second hypothesis, and is likely attributed to the leaching effect and low soil nutrients (Table 3). In contrast to CO_2 fluxes, nutrient addition did not impact CH_4 and N_2O fluxes in this riparian ecosystem, likely owing to the unaltered soil properties, vegetation diversity, and root carbon content (Table 3). This result differs from previous studies demonstrating that increased CH_4 and N_2O emissions under N enrichment but decreased emissions under phosphorus (P) enrichment in grasslands, forests, wetlands, and agricultural ecosystems (Chen et al., 2019, 2020; Martinson et al., 2021; Shen and Zhu, 2022; Wang et al., 2018; Xu et al., 2020). The discrepant results can be attributed to variations in environmental properties, including climate, soil properties, and vegetation characteristics, which influence the uptake, absorption, and availability of nutrients for CH_4 and N_2O production and consumption.

4.3. Implications

There is an uncertainty in predicting global GHG fluxes under climate change owing to the different responses of ecosystems. For example, extreme rainfall increases permafrost thawing, which can accelerate climate warming by releasing carbon from previously frozen soils (Magnusson et al., 2022), while another study has reported that rainfall increase and flooding could reduce CO_2 emission and mitigate climate warming in floodplain because of low aerobic condition and microbial activity (Deng et al., 2018). In this study, we also revealed a mitigation effect of extreme rainfall on climate warming in a riparian ecosystem. Specifically, we observed a significant reduction in CO_2 emissions but an increase in CH_4 and N_2O fluxes following extreme rainfall events (Fig. 1). It should be noted that the stimulation of CH_4 fluxes by extreme rainfall was not observed in the P and N + P treatments, while the stimulation of N_2O fluxes was absent in the PF, N, and N + P treatments. We presumed that the PF, N, and N + P treatments could increase available nutrients for CH_4 and N_2O consumption, which offsets the positive effect of extreme rainfall. These findings underscore the need to collectively consider extreme rainfall events, flooding frequency, and fertilization to avoid overestimating the extreme rainfall effects on N_2O and CH_4 fluxes.

Taking the three GHGs into account, CF significantly reduced the GWP compared with NF, and N addition significantly increased the GWP in the normal rainfall year (Fig. 4). These findings indicate that CF has the potential to alleviate net GHG emissions from riparian ecosystems, whereas N addition can increase the net GHG emissions. Therefore, implementing measures to increase flooding frequency and reduce N input in the reservoir riparian zones would be effective in mitigating GHG emissions. However, the impact of extreme rainfall on GWP outweighs the effects of flooding frequency and nutrient addition, suggesting that extreme rainfall plays a prominent role in regulating GHG fluxes in riparian ecosystems. In addition, extreme rainfall event also significantly reduced the CO_2 proportion in the GWP, which can be attributed to the low diffusion and activity of microbes and vegetation (Altor and Mitsch, 2008). However, the CO_2 remains the major GHG, though the CH_4 proportion increased up to ~40%, suggesting that plant and microbial respiration is the dominant GHG source no matter under which rainfall conditions. It should be noted that we measured ecosystem respiration in this study, and the CO_2 uptake by the vegetation has not been considered. Therefore, further research should take this aspect into account so as to accurately assess whether the riparian ecosystem is a carbon sink or a carbon source.

5. Conclusions

It is of paramount importance to advance our understanding of how GHGs respond to the interactions between climate change and intensified human activities. Our work represents an initial step toward investigating the impacts of extreme rainfall events, flooding frequency, and nutrient addition on GHG fluxes within riparian ecosystems. The findings revealed that extreme rainfall events had a significant effect on GHG fluxes with consequences for understanding climate feedback. Moreover, this study demonstrated that extreme rainfall events overshadowed the influence of flooding frequency and nutrient addition on GHG fluxes because of their overriding impact on soil properties (including soil moisture, pH, dissolved organic carbon, total organic carbon, total carbon, and ammonium content) and vegetation characteristics (such as aboveground biomass and root carbon content). The insights gained from this study have the potential to fine-tune the accuracy of models for predicting GHG emissions in riparian ecosystems and contribute to the development of effective climate mitigation strategies.

CRedit authorship contribution statement

Yu Gong: Writing – original draft, Visualization, Validation, Supervision, Data curation, Conceptualization. **Xiaoling Li:** Writing – review & editing, Validation, Supervision. **Wenxiang Yi:** Writing – original draft, Visualization, Methodology. **Manuel Delgado-Baquerizo:** Writing – review & editing, Supervision. **Guiyao Zhou:** Writing – review & editing, Visualization, Supervision. **Siyue Li:** Writing – review & editing. **Hao Jiang:** Writing – review & editing, Supervision. **Chen Ye:** Writing – review & editing, Validation, Supervision, Funding acquisition. **Quanfa Zhang:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvman.2024.122945>.

Data availability

Data will be made available on request.

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